STANDARD COLLEGE NTUNGAMO
S.5 SUBSIDIARY ICT
MID TERM MARKING GUIDE
S850/1

SECTION A

ITEM 1

(a) ICT Tools Required

Computer Set

A desktop computer system consists of a system unit, monitor, keyboard, and mouse. For Nalule's task, an ideal configuration includes an Intel Core i5 or i7 processor, 16 GB of RAM, a 512 GB Solid State Drive (SSD), running windows 11 professional and dedicated HDMI or DisplayPort outputs. The primary purpose of the computer set is to serve as Nalule's main workstation where she can process digitized historical data, build presentation slides, generate performance charts, and upload files to the web.

Multifunction Printer

A multifunction printer integrates printing, scanning, photocopying, and sometimes faxing into a single physical unit. For school use, it should feature high-resolution color printing (at least 1200 x 1200 DPI), an integrated flatbed scanner, and automatic duplex (two-sided) printing. In this scenario, its built-in scanner is used to digitize the dusty, paper-bound historical archives and photos into computer files, while its printer generates structured, professional color handouts for parents to take home after the induction ceremony.

Digital Projector

A digital projector is an optical output device that receives video signals from a computer and displays the corresponding images onto a projection screen. It requires a minimum brightness rating of 3,500 Lumens, Full HD (1080p) native resolution, and HDMI or wireless connection capabilities. Its purpose in the scenario is to display Nalule's final visual slideshow onto a large screen in the school hall during the Senior One orientation event.

Broadband Modem and Router

A broadband modem establishes an internet connection with the Service Provider, while a router distributes that network connection wired or wirelessly to local devices. Recommended specifications include Dual-Band Wi-Fi, Gigabit Ethernet ports, and high-speed fiber or LTE modem capabilities. This tool provides the network infrastructure necessary for Nalule to access cloud platforms and host a shared online workspace where teachers can review and edit content in real time.

Presentation Software

Presentation software (such as Microsoft PowerPoint or Google Slides) is an application designed to create visual slide sequences. Its key features include pre-designed templates, multimedia embedding (photos, audio, video), slide transition effects, speaker notes, and multi-format export options. In Nalule's scenario, it replaces her plain-text pad by allowing her to organize historical photos, embedded trend charts, and text into a structured, visual slideshow for the induction ceremony.

Spreadsheet Software

Spreadsheet software (such as Microsoft Excel 2024 or Google Sheets) provides interactive grid environments for data organization, mathematical calculations, and analysis. Key features include dynamic data tables, built-in statistical functions (AVERAGE, MAX), and automated chart generation tools (bar, line, and pie charts). Nalule uses this software to input raw academic records from past years and automatically generate trend graphs that illustrate the school's academic performance to guardians and the management committee.

Cloud Storage and Collaboration Software

Cloud storage software (such as Google Drive or Microsoft OneDrive) offers remote file hosting combined with online productivity tools. Key features include real-time multi-user editing, access permissions (view, comment, edit), shareable web links, version history tracking, and automatic sync. Its purpose in the scenario is to provide the shared online workspace requested by the Headteacher, allowing teachers to review, edit, and approve the orientation package simultaneously from their own devices.

Word Processing Software

Word processing software (such as Microsoft Word or Google Docs) provides advanced text formatting, document layout design, image placement, and print configuration. Its key features include typography controls, header and footer styling, page margins, and image wrapping. Nalule uses this tool to organize the digitized historical summaries and performance reports into a clean, well-formatted layout for the physical handout parents take home.

(b) Analysis & Troubleshooting of the Blank Monitor Display

The computer system unit powers on indicated by fan noise or system lights, but the monitor screen remains totally black. This behavior points to a breakdown in signal transmission or power delivery to the display. Common root causes include a loose display cable, an unpowered monitor, an incorrect input source selection on the monitor, static charge buildup inside the system unit, or an internal hardware failure such as improperly seated RAM modules or a faulty graphics card.

Verify Monitor Power Supply and Indicator LED:

Observe the monitor's power indicator light. If the light is completely unlit, verify that the power cord is firmly seated in both the monitor inlet and the wall outlet. Ensure the wall switch is turned on and press the monitor's power button directly.

Inspect and Reseat Video Signal Cables:

Locate the video cable (HDMI or VGA) connecting the monitor to the system unit. Unplug both ends and reconnect them firmly, tightening any thumb screws on VGA cables. If the system unit has a dedicated graphics card installed, ensure the cable is plugged into the graphics card port rather than the motherboard's integrated port.

Select the Correct Video Input Source:

Press the **Input** or **Source** button located on the monitor's physical menu controls. Cycle through the available display channels (e.g., HDMI 1, HDMI 2, VGA) until it matches the specific port and cable connected to the computer.

Perform a Hard Power Cycle and Static Discharge:

Shut down the system unit completely, switch off the main wall outlet, and disconnect the power cable from the back of the tower. Press and hold the system unit's power button for 15 to 20 seconds to drain residual electricity from the internal components. Reattach the power cable and turn the computer back on.

Cross-Test Hardware with Alternative Devices:

Connect the monitor to another working laptop or desktop using a functional video cable. Alternatively, connect Nalule's system unit to a different working monitor or the school projector. If the original monitor works on another device, her original display cable or monitor settings were at fault. If the screen remains black on a second monitor, the issue lies inside the system unit's internal hardware (such as unseated RAM or graphics card).

SECTION B

ITEM 2

(a) Classification and Application of Hardware Tools

To resolve Muhindo's operational challenges, his required hardware tools are categorized below by their functional system roles:

Input / Media Capture Devices

Input devices are physical peripherals used to enter raw data, instructions, and commands into a computer system. They convert human-understandable actions or environmental signals (such as keypresses, physical movements, or optical audio/visual signals) into digital data that the processing unit can interpret.

Examples of input devices

- **Camcorder / Digital Camera:** Captures high-quality video clips and high-resolution still images of agricultural activities and community events, replacing scatter-shot capture methods.

- **Studio Condenser Microphone:** Records clear, studio-grade audio tracks and interviews with local elders without background noise, providing pristine sound for his documentaries and folk music recordings.
- **Keyboard and Mouse:** Primary human-interface devices used to navigate complex media editing applications, enter text for scripts, and control software operations precise to fractions of a second.

Processing Hardware (System Unit Components)

Processing devices form the internal computing core of the computer, responsible for executing software instructions, performing mathematical computations, managing data flow, and transforming raw input into meaningful information. The Central Processing Unit (CPU) acts as the primary brain of the system, while specialized processors like the Graphics Processing Unit (GPU) handle complex graphical computations.

Examples of processing devices

- **High-Performance Multi-Core CPU (e.g., Intel Core i7/i9 or AMD Ryzen 7/9):** Provides the computational core needed to process complex media editing projects and render heavy audio-visual files.
- **Dedicated Graphics Processing Unit (GPU):** Accelerates video rendering, timeline playback, and graphical processing, enabling fluid editing of high-definition video footage.

Output Devices

Output devices receive processed data from the computer system and translate it into a human-readable or usable physical format, such as visual displays, sound waves, or printed documents. They communicate the results of processing back to the user or an audience.

Examples of output devices

- **High-Lumen Digital Projector:** Displays educational slides, workshops, and documentary films onto a large wall or screen so an audience of fifty villagers can watch comfortably together.
- **Desktop Color Printer / Multifunction Printer:** Produces physical, hard-copy training manuals and completion certificates for youths attending his media workshops.
- **Desktop Monitor:** Acts as Muhindo's primary workstation display for precise visual feedback during video editing, script formatting, and file management.

Communication & Networking Hardware

Communication devices facilitate the transmission and reception of digital data between two or more computer systems over local networks or the internet. They establish physical or wireless connections to allow resource sharing, web browsing, cloud backups, and remote file distribution.

Examples of communication devices

- **Broadband Modem and Wi-Fi Router Setup (or Cellular 4G/5G Gateway):** Establishes a stable broadband internet connection directly in his village office, eliminating the need to travel physically to deliver media files to city distributors or upload work to online platforms.

Storage Hardware

Storage devices are physical components used to hold data, software programs, and files either temporarily during active processing or permanently for long-term retention. Primary storage (like RAM) holds active data for fast CPU access, while secondary storage (such as SSDs and external HDDs) keeps files permanently without requiring continuous power.

Examples of storage devices

- **High-Capacity External Hard Drives (HDDs)/Magnetic Tape:** Provides terabytes of secure, long-term storage to archive raw historical footage off the main computer, preventing workspace clutter and protecting assets against system crashes or hardware failure.

(b) Physical Assembly Procedure for the Desktop Workspace

Unpack and Prepare the Workspace

Unbox all components and place the system unit (tower), monitor, and peripherals on a clean, stable surface. Ensure the power switch on the back of the system unit is flipped to the "OFF" (0) position.

Connect the Monitor to the System Unit

Place the monitor in the center of the desk. Connect one end of the display cable (HDMI or DisplayPort) to the video port on the back of the monitor, and plug the other end directly into the dedicated graphics card port on the back of the system unit tower (not the motherboard port). Tighten any retaining screws if using a VGA/HDMI cable.

Attach Input Peripherals (Keyboard and Mouse)

Plug the USB cables from the keyboard and mouse into the rear USB ports on the system unit. Using the rear ports reserves the convenient front USB ports for temporary devices like memory cards, cameras, and external hard drives.

Connect Specialized Media Hardware

Plug the studio condenser microphone into an audio interface or dedicated USB port on the system unit. Connect the printer to a USB port or network port on the router. Connect the external archival hard drive to a high-speed USB 3.0/USB-C port on the tower.

Connect the Network Cable

Run an Ethernet network cable from the LAN port of the internet router/modem directly into the Ethernet (RJ-45) port on the back of the system unit for a stable, high-speed connection.

Connect Power Cables to an Uninterruptible Power Supply (UPS)

Plug the power cables for the system unit, monitor, projector, and printer into a Uninterruptible Power Supply (UPS) or surge protector. *Do not plug the equipment directly into wall sockets to protect against village voltage spikes or sudden outages.* Plug the UPS into the main electrical wall outlet.

(C) Sequential Daily Power-On Procedure

Perform Physical Connection and Cable Inspection:

Before turning on any electrical switches, physically inspect all workstation connections. Ensure display cables (HDMI/DisplayPort) are firmly seated in both the monitor and graphics card, USB cables for peripherals and external hard drives are securely plugged in, and audio cables are firmly connected. Check that all power cables are completely pushed into their respective sockets on the system unit, monitor, printer, and wall outlets, ensuring no wires are loose, pinched, or damaged.

Switch On the Main Wall Outlet and UPS:

Turn on the primary power outlet at the wall, then switch on the Uninterruptible Power Supply (UPS). Allow the UPS to initialize until its power LED stabilizes and it stops beep-checking. This conditions incoming power and protects connected gear against line noise or voltage spikes.

Turn On Networking & External Peripherals First:

Switch on the internet modem/router, studio microphone interface, external hard drives, printer, and digital projector. Turning on external devices first ensures the operating system automatically detects and mounts them when the computer boots.

Turn On the Desktop Monitor:

Press the power button on the primary computer monitor so it is actively listening for an incoming display signal before the computer initializes its graphics hardware.

Power On the System Unit Tower:

Finally, press the main power button on the front or top of the desktop computer tower. The system unit will execute its Power-On Self-Test (POST), initialize internal components smoothly, and boot into the operating system.

ITEM 3

(a) Assessment of Ateenyi's Storage Challenges and File Management Best Practices

Ateenyi is experiencing serious file management problems caused by bad storage habits. Storing all files directly on the desktop slows down computer startup performance, increases the risk of file loss during operating system crashes, makes file retrieval slow, and causes accidental file overwriting due to vague, non-descriptive file names.

To fix these challenges, Ateenyi should implement the following best practices for file and folder management:

Adopt a Logical Folder Hierarchy:

Instead of saving files directly on the desktop, Ateenyi should create a dedicated root directory in the Documents or Local Disk drive (e.g., Youth_Empowerment_Summit_2026). Inside this folder, he should create subfolders based on file categories or departments, such as:

- 01_Registration_Forms
- 02_Budgets_and_Finance
- 03_Guest_Speakers
- 04_Project_Videos

Use Clear, Descriptive File Naming Conventions: Files should be saved using standardized, informative names that describe their content, author, or status instead of default names like "Document1" or "Final Copy". For example:

- Budget_Youth_Summit_Audited_v1.xlsx
- Profile_Hon_Speaker_John.docx
- Video_Youth_Agri_Project_2026.mp4

Implement Version Control to Avoid Overwriting: To prevent overwriting critical documents, Ateenyi should append version numbers (e.g., _v1, _v2, _Draft, _Approved) or dates (e.g., YYYY-MM-DD) at the end of file names whenever updates are made, rather than using generic tags like "Final Copy".

Regular Cleanup and Archiving: Unused shortcuts, temporary files, and obsolete drafts should be deleted or moved to an Archive folder. Keeping the computer desktop clean improves overall system speed and visual clarity.

Maintain Data Backup Systems: Important summit records should be backed up regularly to secondary storage devices (such as a USB flash drive or external hard drive) or cloud storage platforms (such as Google Drive or OneDrive) to protect against unexpected hardware failure.

(b) Data Size Analysis, Calculations, and Flash Disk Recommendation

To determine the exact capacity needed, all file sizes are converted to Gigabytes (GB) using standard binary computer storage conversions (1 GB = 1,024 MB and 1 MB = 1,024 KB).

Total Size of Word Documents

- Number of Word documents = 120 files
- Average size per file = 5 MB
- Total size in Megabytes (MB) = Number of Word documents x Average size per file
- Total size in Megabytes (MB) = 120 × 5 MB = 600 MB

- Conversion to Gigabytes (GB) = 600 MB ÷ 1,024 MB/GB = 0.586 GB (or 0.600 GB in decimal standard)

Total Size of Excel Spreadsheets

- Number of Excel spreadsheets = 40 files
- Average size per file = 2,048 KB
- Conversion per file to MB = 2,048 KB ÷ 1,024 KB/MB = 2 MB
- Total size in Megabytes (MB) = Number of Excel spreadsheets x Conversion per file to MB
- Total size in Megabytes (MB) = 40 × 2 MB = 80 MB
- Conversion to Gigabytes (GB) = 80 MB ÷ 1,024 MB/GB = 0.078 GB (or 0.080 GB in decimal standard)

Total Size of High-Definition Video Clips

- Number of video clips = 15 clips
- Exact size per clip = 1.2 GB
- Total size in Gigabytes (GB) = Number of video clips x Exact size per clip
- Total size in Gigabytes (GB) = 15 × 1.2 GB = 18.000 GB

Total Data Capacity Required

- Total size of files = Size of Word Docs + Size of Spreadsheets + Size of Videos
- Total size of files = 0.586 GB + 0.078 GB + 18.000 GB = 18.664 GB

Add Required Free Space Buffer

Required free space buffer = 2.000 GB

Total required storage capacity = Total size of files + Required free space buffer

Total required storage capacity = 18.664 GB + 2.000 GB = 20.664 GB

Conclusion & Recommendation

Ateenyi requires a minimum storage capacity of 20.664 GB to store all summit files safely while maintaining his desired 2 GB free space.

Commercial flash drives are manufactured in standard capacities of 4 GB, 8 GB, 16 GB, 32 GB, 64 GB, and 128 GB. Since 16 GB is insufficient to hold the 20.664 GB total, Ateenyi must purchase a 32 GB flash disk. This is the most cost-effective option that comfortably fits all his files and leaves the required space for future updates.

END